

Internet Gateway

Bidirectional, Highly Available Internet Connectivity for a VPC

Interview Preparation Guide

Problem & motivation - 1:1 NAT internals - public IP requirement - packet flow - failure modes - scaling - security - observability & automation - cost - troubleshooting - real-world patterns

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Executive Summary

An Internet Gateway (IGW) is the VPC component that enables resources with public IP addresses to communicate with the internet in both directions. It is a managed, horizontally scaled, redundant target - not an instance - so it imposes no bandwidth bottleneck and no availability risk. The IGW performs one-to-one network address translation between an instance's public/elastic IP and its private IP. Crucially, an IGW alone is not enough: a subnet needs a 0.0.0.0/0 route to the IGW AND the instance needs a public IP. This guide covers internals, packet flow, failure modes, scaling, security, observability, automation, cost, and troubleshooting.

1. The Problem It Solves

Instances in a VPC use private RFC 1918 addresses that are not routable on the public internet. Something must bridge that private space to the internet, scale to any traffic level, and stay highly available. The IGW is that bridge for inbound and outbound public traffic.

Core needs the IGW addresses

- **Public reachability:** Let internet clients reach public-facing services (web servers, load balancers).
- **Outbound for public hosts:** Let instances with public IPs initiate internet connections.
- **Address translation:** Map private IPs <-> public/elastic IPs without per-instance NAT software.
- **No bottleneck:** Provide internet connectivity without a throughput or availability single point of failure.

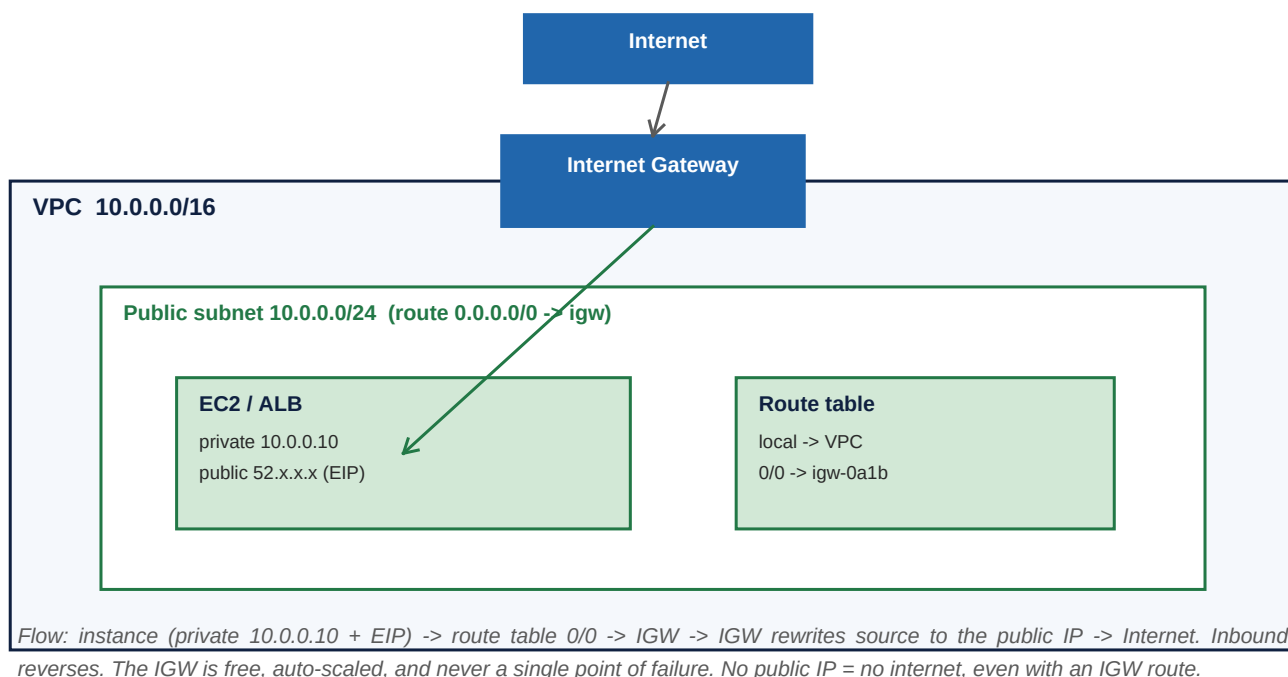
2. How It Works Internally

An IGW is attached to a VPC (one IGW per VPC). It is implemented in the AWS network fabric as a horizontally scaled, redundant service rather than an EC2 instance, so it has no managed bandwidth limit and no maintenance windows. For traffic to use it, the source subnet's route table must point 0.0.0.0/0 (or specific public CIDRs) at the IGW.

- It performs 1:1 NAT: outbound, it rewrites the instance's private source IP to its associated public/elastic IP; inbound, it does the reverse.
- If an instance has only a private IP (no public/EIP), the IGW cannot give it internet access - it has nothing to translate to.
- IGW is stateless from the customer's perspective; security is enforced by security groups and NACLs.
- For IPv6, an egress-only Internet Gateway provides outbound-only access.

Reference Diagram - Internet Gateway data path

An Internet Gateway (IGW) is a horizontally scaled, redundant, VPC-attached component that performs 1:1 NAT between an instance's public IP and its private IP, enabling bidirectional internet connectivity for resources in public subnets.



3. Traffic Flow, Packet Flow & Routing Decisions

Outbound packet via IGW

- Instance (private 10.0.0.10, public 52.x.x.x) sends to an internet host.
- Security group egress is evaluated (stateful).
- Route table: 0.0.0.0/0 -> igw target.
- IGW rewrites the source IP from private to the instance's public IP.
- Packet exits to the internet; replies return to the public IP and the IGW maps them back to the private IP.

The #1 IGW gotcha

An IGW route is necessary but NOT sufficient. The instance must also have a public IPv4 or Elastic IP. A private-only instance in a 'public' subnet still cannot reach the internet through an IGW - it needs NAT instead.

4. Advantages & Disadvantages

Advantages

- Fully managed, horizontally scaled, and redundant - no SPOF.
- No hourly charge for the gateway itself.
- Supports both inbound and outbound public traffic.
- No bandwidth ceiling to manage.

Disadvantages / costs

- Exposes resources publicly if misused - requires disciplined SG/NACL/routing.
- Only useful for resources that have public IPs.
- One IGW per VPC; cannot segment internet egress within a VPC by IGW.
- Standard internet data-transfer-out charges still apply.

5. Common Failure Modes

Failure	Symptom	Typical root cause
IGW not attached	No internet at all	IGW created but not attached to the VPC.
No 0/0 route	Timeouts	Subnet route table missing default route to IGW.
No public IP	Outbound fails	Instance has only a private IP; needs EIP or NAT.
SG/NACL block	Refused/one-way	Security group or NACL denies the port/return traffic.
Public IP not assigned	Intermittent after restart	Auto-assign public IP off; ephemeral IP lost on stop/start.

6. Scaling Considerations

- **Throughput:** IGW scales automatically; no action needed for bandwidth.
- **HA:** IGW is inherently redundant - nothing to make multi-AZ.
- **Egress design:** For many private hosts, pair public subnets (IGW) with NAT for private subnets.
- **EIP limits:** Public IPv4 / EIP quotas can bound how many public hosts you run; request increases as needed.

7. Security Implications

- An IGW route turns a subnet public - apply it deliberately and narrowly.
- Lock down inbound with security groups (default deny) and NACLs.
- Prefer load balancers in public subnets and keep app/data tiers private.
- Use VPC Flow Logs and GuardDuty to watch for exposure and abuse.
- Consider AWS Network Firewall / WAF in front of internet-facing services.

8. Observability, Reliability & Automation

Observability

- Flow Logs to see internet-bound accepts/rejects.
- CloudWatch/metrics on the public-facing resources (ALB/EC2), not the IGW itself (the IGW exposes no per-gateway metrics).

Reliability

- IGW is managed and redundant; reliability work focuses on your resources.
- Spread public-facing resources (ALB targets) across AZs.

Automation

- Create + attach IGW and add routes in IaC.
- Guardrails to flag new 0/0->IGW routes on sensitive subnets.

9. Cost Considerations

Cost driver	Why it adds up	Mitigation
IGW gateway	No hourly/processing charge for the IGW.	N/A - the IGW is free.
Data transfer out	Internet egress (GB out) is billed.	CDN/caching, compression, fewer external calls.
Public IPv4	Public/Elastic IPv4 addresses now carry an hourly charge.	Release unused EIPs; consolidate behind load balancers; use IPv6.

10. Troubleshooting Approach

For 'instance can't reach the internet':

- 1. Is an IGW attached to the VPC?
- 2. Does the subnet's route table have 0.0.0.0/0 -> igw?
- 3. Does the instance have a public/Elastic IP?
- 4. Do SG (egress) and NACL (both directions) allow the traffic?
- 5. Confirm with Reachability Analyzer + Flow Logs.

Mnemonic

'Attached -> Route -> Public IP -> SG/NACL.' If the host has no public IP, an IGW will never help - use a NAT Gateway.

11. Real-World Examples & Patterns

- **Public ALB:** Internet-facing load balancer in public subnets fronting private app servers.
- **Bastion (legacy):** A jump host with an EIP in a public subnet (prefer SSM Session Manager instead).
- **Egress-only IPv6:** Outbound-only internet for IPv6 private workloads via egress-only IGW.
- **Public + NAT combo:** Public subnets host IGW-facing LB/NAT; private subnets reach internet via NAT.

12. Senior-Level Trade-offs & Talking Points

- Public IP per instance vs load balancer + private fleet: simplicity vs security and IPv4 cost.
- IGW (bidirectional) vs NAT (outbound-only) vs egress-only IGW (IPv6): pick by direction and IP family.
- Single IGW per VPC: simple but no per-IGW egress segmentation - use routing and inspection instead.
- IPv4 cost pressure vs IPv6 adoption readiness.

Whiteboard checklist

(1) Draw VPC + attached IGW. (2) Public subnet with 0/0->IGW. (3) Instance with public IP. (4) Show 1:1 NAT out and back. (5) Contrast with NAT GW for private hosts. (6) Discuss SG/NACL, IPv4 cost, and exposure risk.

13. Quick Glossary

Term	Meaning
IGW	Internet Gateway - bidirectional internet access for public IPs.
Egress-only IGW	Outbound-only internet access for IPv6.
EIP	Elastic IP - a static public IPv4 address you own.
1:1 NAT	Mapping one private IP to one public IP (what the IGW does).
Public subnet	Subnet with a 0/0 route to an IGW.
Auto-assign public IP	Subnet setting granting ephemeral public IPv4.
DTO	Data Transfer Out - billed internet egress.